Lab 7

Week 8-9

Isolated DC-DC Converters: Simulation

Objectives

- Identify the switching intervals and energy-transfer paths of flyback and forward converters.
- Simulate and measure isolated DC-DC converter voltage and current waveforms in LTspice.
- Distinguish flyback CCM and DCM, and verify transformer reset in a forward converter.

Equipment and components

- Computer with LTspice installed.

Preliminary

- Review the Isolated DC-DC Converters lecture: transformer dot convention, magnetizing inductance, volt-second balance, and converter voltage gains.
- Review LTspice coupled inductors, waveform plotting, cursors, and average measurements.

Procedure

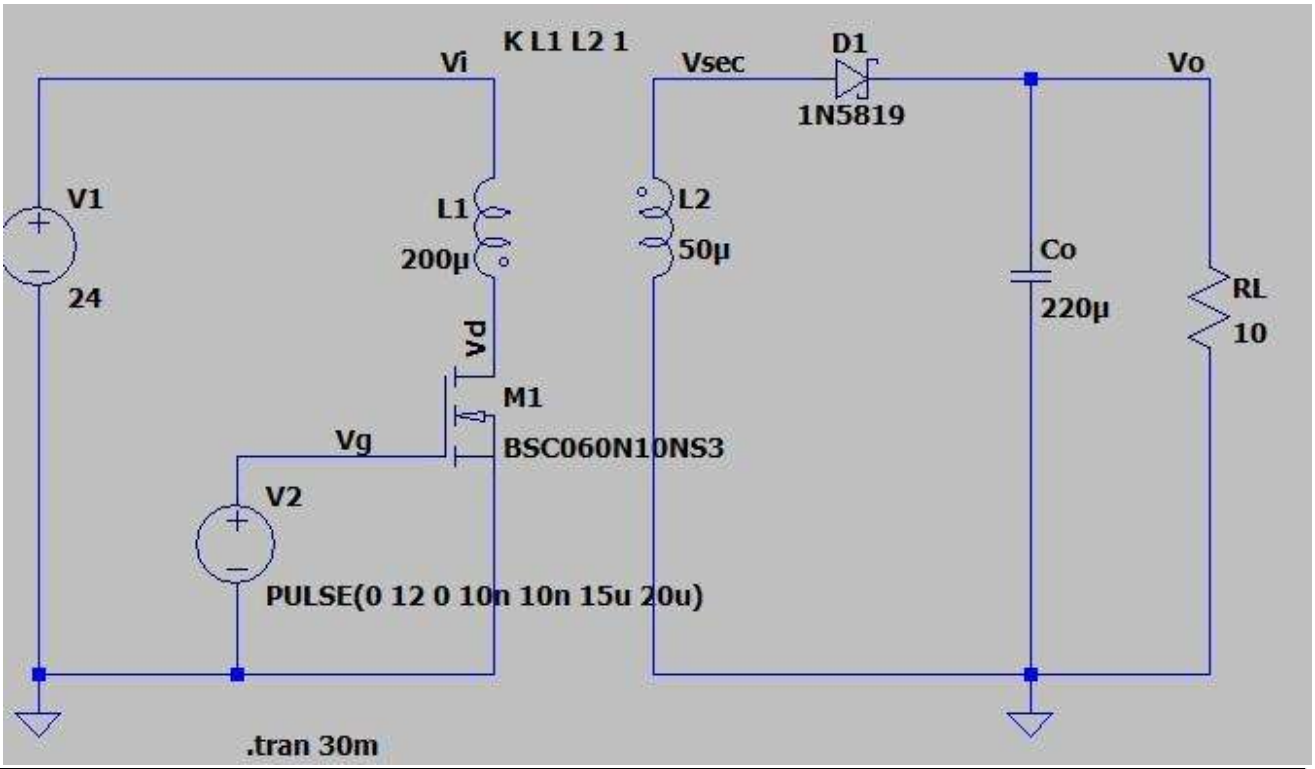
Part A: Flyback Converter in Continuous Conduction Mode (CCM)
Part B: Flyback Converter in Discontinuous Conduction Mode (DCM)
Part C: Forward Converter

Name: _Nathan Zimmerman, Ashton Capron, Roberto Macias, Matt Lezhansky

Date: _____10-19-26_____

Part A: Flyback Converter in CCM

1. Launch LTspice and create a new schematic. Construct the Flyback circuit below.



Flyback Parameter	Value
Input voltage V1	24 V
Primary inductance $L_1 = L_m$	200 µH
Secondary inductance L_2	50 µH
Output capacitor C_o / load R_L	220 µF / 10 Ω
Coupling coefficient	K = 1 (Set by Spice Directive .t)
Gate pulse V2	PULSE (0 10 0 10n 10n 10u 20u)

Notes:

For MOSFET M1, select component "nmos"; right-click and select BSC060N10NS3, (or any MOSFET Vds[V] ≥ 100*V*);

For Diodes, right-click and select 1N5819;

For the two coupled inductor L1 and L2, in move mode → select L1 or L2→ Ctrl+E to mirror the L1 or L2. → Right-Click L1 and L2, turn on 'Show Phase Dot' and check the winding dots position for them.

To make L1 and L2 coupled, Click SPICE Directive (S) → Type 'K L1

L2 1' and drag the text on the schematic;

Click Label Net, and add the voltage nodes: Vi, Vo, Vsec, Vd and Vg as shown in the circuit diagram.

Use $n = \frac{N_2}{N_1}$ throughout this lab.

$$n = \frac{N_2}{N_1} \sqrt{\frac{L_2}{L_1}} \qquad T_s = \frac{1}{f_s}$$

From the setting above:

Switching period T_s = ____20_____ µs. Switching Frequency f_s = ___50000_____ Hz;

Duty Cycle d = ___0.5_____. Turns ratio $n = \frac{N_2}{N_1}$ = ___0.5_____;

Save your schematic as Lab7_A_Flyback.asc.

2. Run the Transient (Stop Time: 30m)

In the plot window, add 6 panes (Right-click and Add Plot Pane Below):

1) Plot the gate-source voltage V(Vg)

2) Plot the output voltage V(Vo).

3) Plot the primary winding voltage V(Vi,Vd).

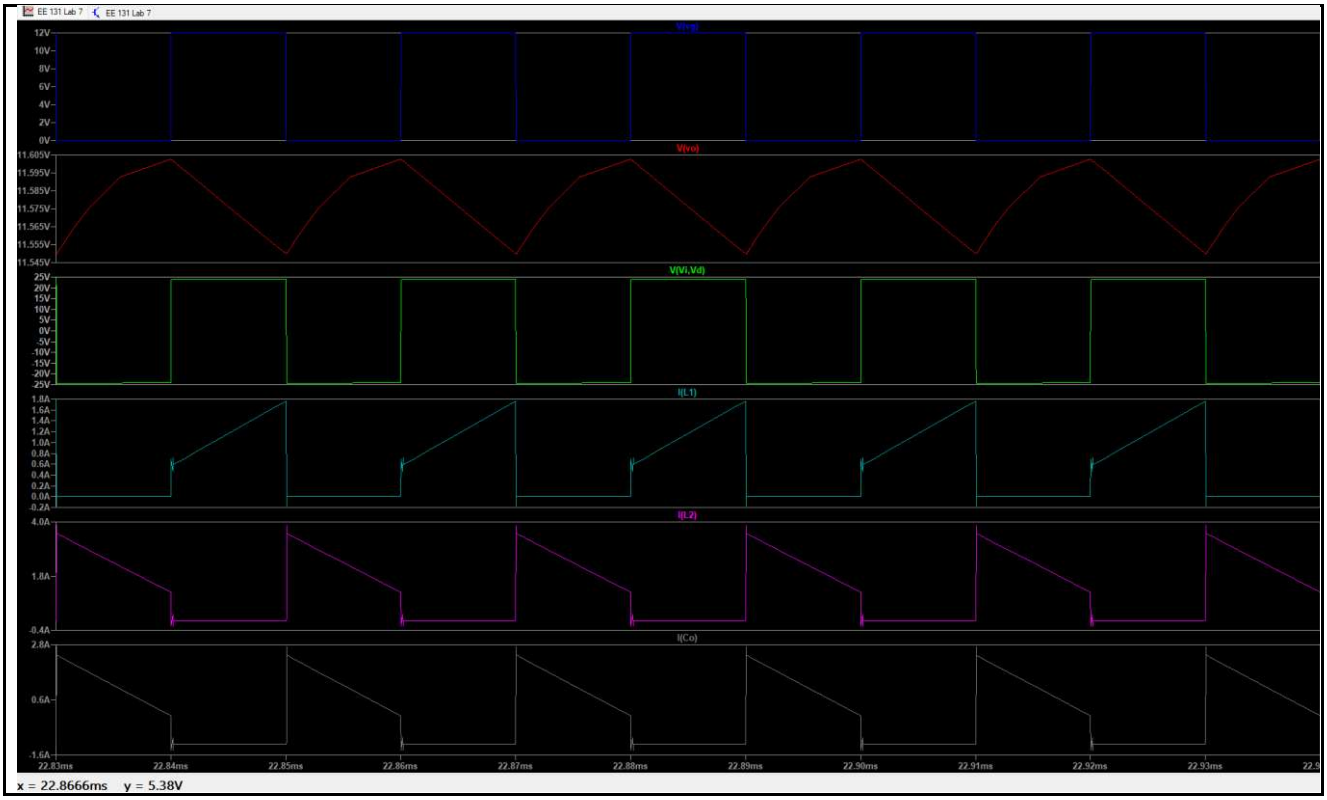
Hint: To measure the voltage difference between two floating nodes, move the cursor to the first node Vi until the voltage probe appears. Then, hold the left mouse button and drag the probe from the first node to the second node Vd

4) Plot the primary winding current I(L1).

5) 5) Plot the secondary winding current I(L2)

6) 6) Plot the output capacitor current I(Co).

Show your CCM waveforms screenshot:



3. Complete the table using your waveforms. Use voltage levels away from the switching edges.

Quantity	Switch ON	Switch OFF
D1 conducting? (Yes/No)	No	Yes
Primary voltage V(Vi,Vd)	23.96	-24.23
Magnetizing current rises or falls?	rises	falls
Element supplying the load	Co	L2
Energy stored or released by L_m?	stored	released

Use the ideal CCM equations below. V_o and I_o are cycle-average output values; Δi_m and ΔV_o are peak-to-peak ripple values.

$$V_o = \frac{1}{n} V_{in} d \qquad I_o = \frac{V_o}{R}$$

$$\Delta i_m = \frac{V_{in} d}{1-d} \qquad I_{m,avg} = \frac{n d I_o}{L_m f_s}$$

$$I_{m,min} = I_{m,avg} - \frac{\Delta i_m}{2} \quad \Delta V_o \approx \frac{I_{oo} f_s}{C}$$

CCM requires the reconstructed magnetizing current to remain positive throughout the cycle.

Quantity	Calculated	Simulated
Average output voltage V_o (V)	12	11.58
Output ripple ΔV_o (V, p-p)	.54V	51.53 mV
Magnetizing ripple Δi_m (A, p-p)	1.2	1.749
Minimum magnetizing current (A)	0	156.49 u

Q1: During the MOSFET ON interval, where is energy stored in the flyback converter, and which component supplies the load?

When the MOSFET is on, the inductor stores the energy in the flyback convertor while the output capacitor supplies the load.

Q2: After the MOSFET turns OFF, how does the stored energy reach the output? Use the primary and secondary current waveforms to explain the energy-transfer sequence.

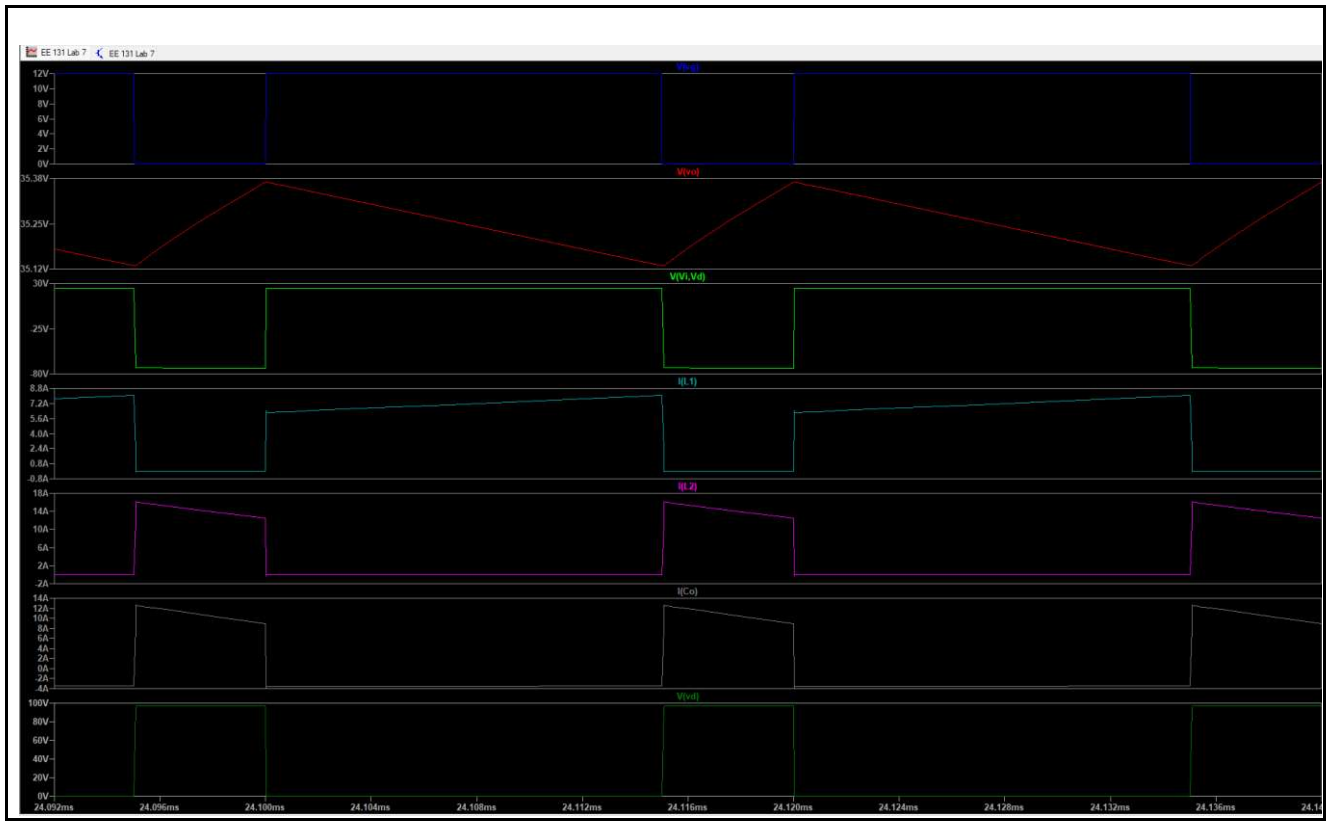
When the MOSFET turns off, the reversed polarity voltage goes through the diode and the stored energy goes into the output capacitor.

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4. Change the Duty Cycle d to 0.75.

Change the pulse ON time from 10u to 15u. Keep the period at 20u and R_L at 10 Ω. Run the transient again and plot the same 6 panes, and add one more pane for the Drain Voltage V(Vd) below V (Vi, Vd)

Show your waveform screenshot at d = 0.75:



Expected output voltage V_o = ___36_____ V.

Simulated average output voltage V_o = _35.245_____ V.

MOSFET Voltage Stress during OFF time: $V_{D,off} = V_{in} + Vo/n$

Calculated $V_{D,of}$ = _____96_____ V; Simulated $V_{D,off}$ = __94.49_____ V.

Q: How do the output voltage and MOSFET voltage stress change as the duty cycle d increases? What should you consider when selecting the MOSFET parameter from datasheet for a flyback converter?

The output Voltage and voltage stress increases as duty increases. When selecting the MOSFET parameter from a datasheet for a flyback converter I would look at the voltage ratings for the MOSFET to make sure that it can handle the increased output and voltage stress.

Part B: Flyback Converter in DCM

1. Restore d = 0.5 (pulse ON time = 10u). Keep all other Part A values, then increase R_L to 50 Ω.

Increasing R_L can produce a demagnetized interval after secondary energy transfer ends. MOSFET capacitances may cause small current ringing during this interval.

First estimate the CCM/DCM boundary using the ideal relationships:

$$I_{o,crit} = \frac{(1-d)\Delta i^{m}}{2n}$$

$$R_{crit} = \left(\frac{2}{1n^{2}} - L_m\, df\right)_{s2}$$

Calculate: Δi_m = ____.6_____ A; $I_{o,crit}$ = ____20_____ A.

Calculate: R_{crit} = 18.18 Ω.

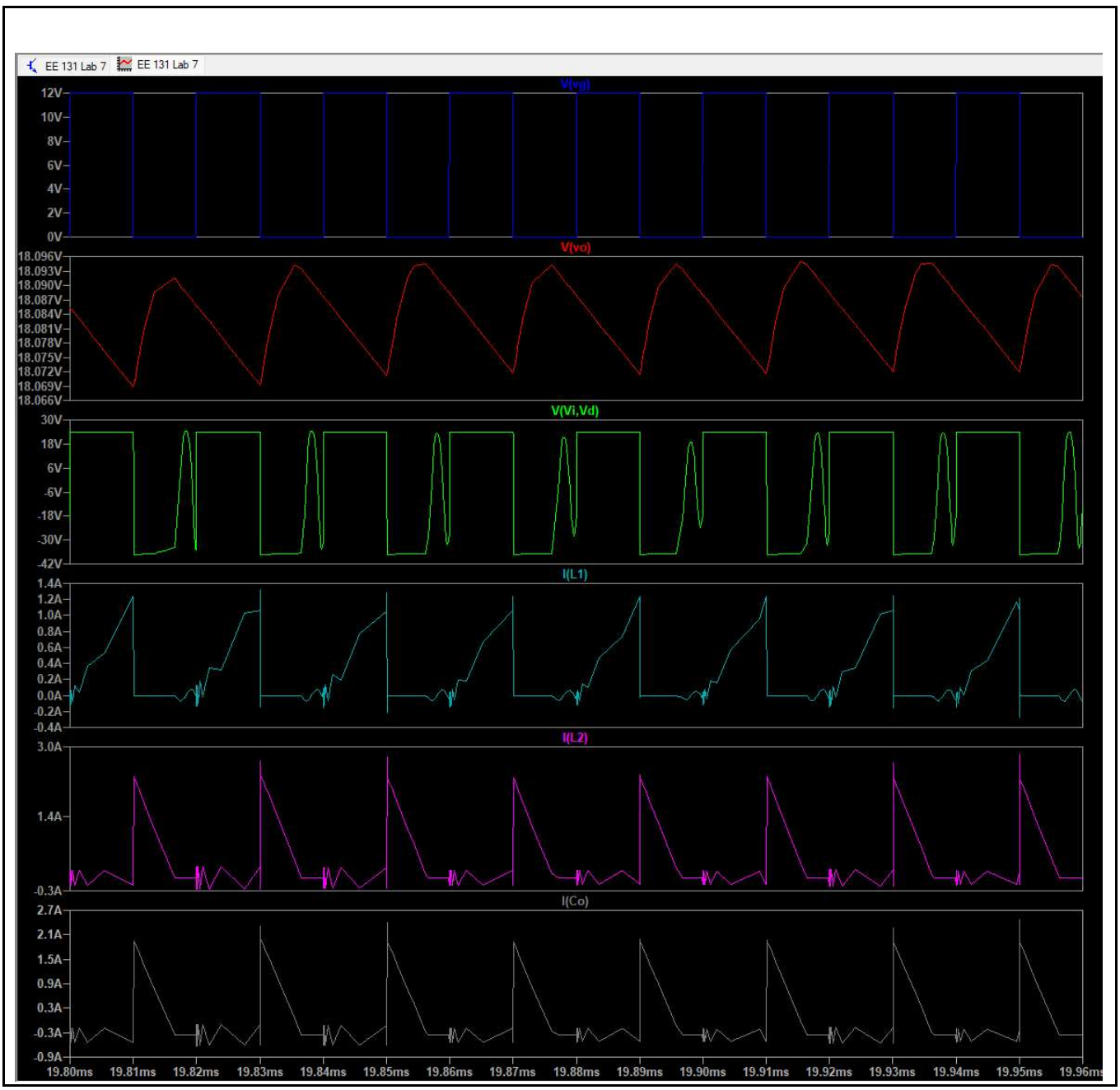
For the same input, duty cycle, and inductances: $R_L < R_{crit}$ gives CCM; $R_L > R_{crit}$ gives DCM in the ideal model. Device drops shift the exact boundary.

2. Run the Transient (Stop Time: 30m).

Plot the same 6 panes as Part A. Zoom in to show the three intervals in one switching period:

1) **M1 ON; magnetizing current rises.**
2) **M1 OFF and D1 conducting; magnetizing current falls**
3) **M1 and D1 OFF; the core is demagnetized, with possible small current ringing.**

Show your DCM waveform screenshot (R_L = 50 Ω):



4. Calculate the ideal DCM output voltage and conduction intervals.

For DCM, the stored magnetizing energy is transferred to the output each cycle. Assume small output ripple and neglect losses.

$$I_{m,pk} = \frac{V_{in} d}{L_m f_s}$$

$$V_o = V_{in} d \sqrt{\frac{R_L}{2 L_m f_s}}$$

$$\delta = \frac{nV_{in}d}{V_o} \qquad t_{zero} = (1 - d - \delta)T_s$$

Here, δT_s is the diode conduction time. Check $d + \delta < 1$. In the ideal model, t_{zero} is the demagnetized interval. Although n cancels from the voltage expression, it sets the conduction interval and mode boundary.

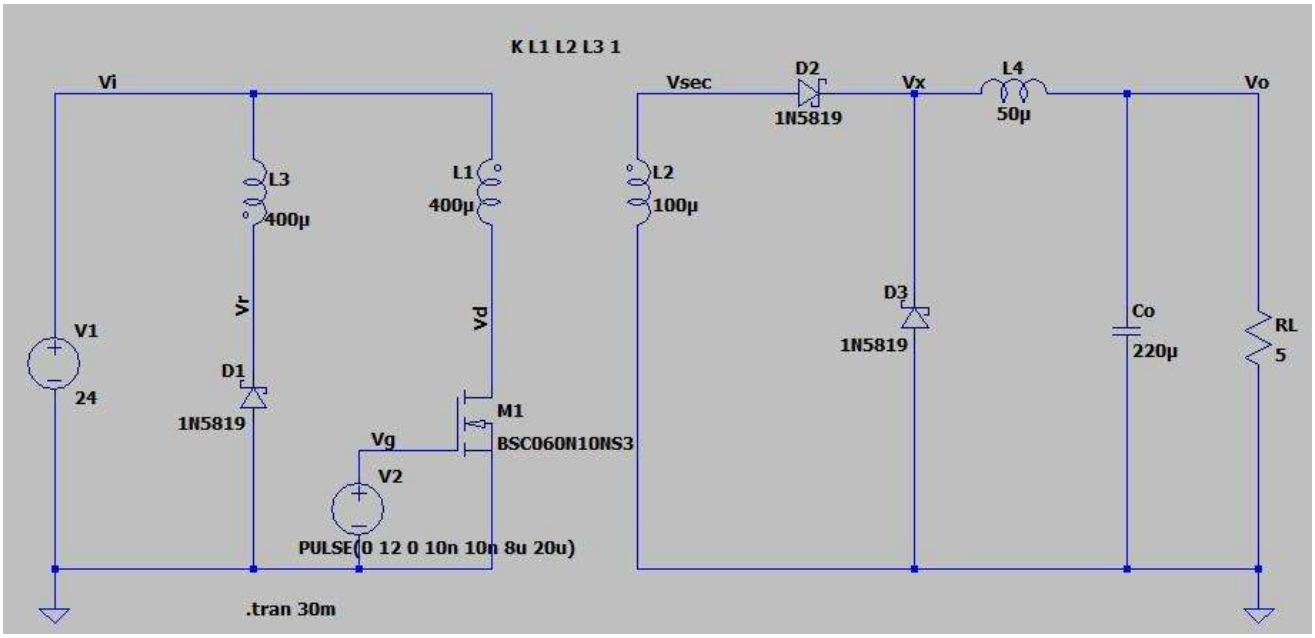
Quantity (R_L = 50 Ω)	**Calculated**	**Simulated**
V_o (V)	18.97	18.084
Diode conduction fraction δ	0.3015	0.3318
Zero-current interval t_{zero} (μs)	3.97	3.364

Q: With d and f_s fixed, predict how V_o changes if R_L increases further.

If RL increases further, Vo will increase with it. This is because RL is in the numerator of the function for Vo, meaning they have a proportional relationship.

Part C: Forward Converter

1. Create a new schematic. Construct the single-switch Forward converter below, including the reset winding L_r.



2. Set the component values and the gate signal.

Forward Parameter	**Value**
Input voltage V1	24 V
L_1 / L_2 / L_3	400µH / 100 µH / 400µH
Turns ratios	$n = \frac{N_2}{N_1} = 0.5$; $\frac{N_1}{N_3} = 1$
Output inductor L_4	50 µH;
Output capacitor C_o / load R_L	220 µF / 5 Ω
Transformer model	K = 1;
Gate pulse V2	PULSE (0 10 0 10n 10n 8u 20u)

Notes:

For MOSFET M1, select component "nmos"; right-click and select

BSC060N10NS3, (or any MOSFET Vds[V] ≥ 100*V*);

For Diodes, right-click and select 1N5819;

For the three coupled inductor L1, L2, and L3, in move mode → select L1/ L2/ L3→ Ctrl+E to mirror the L1/ L2/ L3. → Right-Click L1/ L2/ L3, turn on 'Show Phase Dot' and check the winding dots position for them.

To make L1, L2 and L3 coupled, Click SPICE Directi → Type 'K L1 L2 L3 1' and drag the text on the schematic;

Click Label Net , and add the voltage nodes: Vi, Vo, Vsec, Vx, Vd, Vr and Vg as shown in the circuit diagram.

d

Use one coupling statement for all three windings. Check $= \sqrt{\frac{L_2}{L_1}}$ $\frac{3}{1} = \sqrt{\frac{L_3}{L_1}}\ n$ and N .

N

$$N^3$$

$$t_{\text{reset}} = N__1\, dT_s$$

$$d + \frac{t_{\text{reset}} N_1}{T_s} \leq 1 \qquad d \leq \frac{\quad}{N_1 + N_3}$$

Calculate: f_s = _50,000 Hz_ Hz; d = _0.4_; t_{reset} = _8_ µs.

For $N_1 = N_3$, the theoretical maximum duty cycle is _0.5_.

Use d = 0.4 to leave reset margin. Do not repeat the flyback duty-cycle increase in this part. Save as Lab7_C_Forward.asc.

3. Run the Transient (Stop Time: 30m).

In the plot window, add and plot 9 panes:

1) Gate-source voltage V(Vg) 2) Drain-source voltage V(Vd).

3) Output voltage V(Vo).

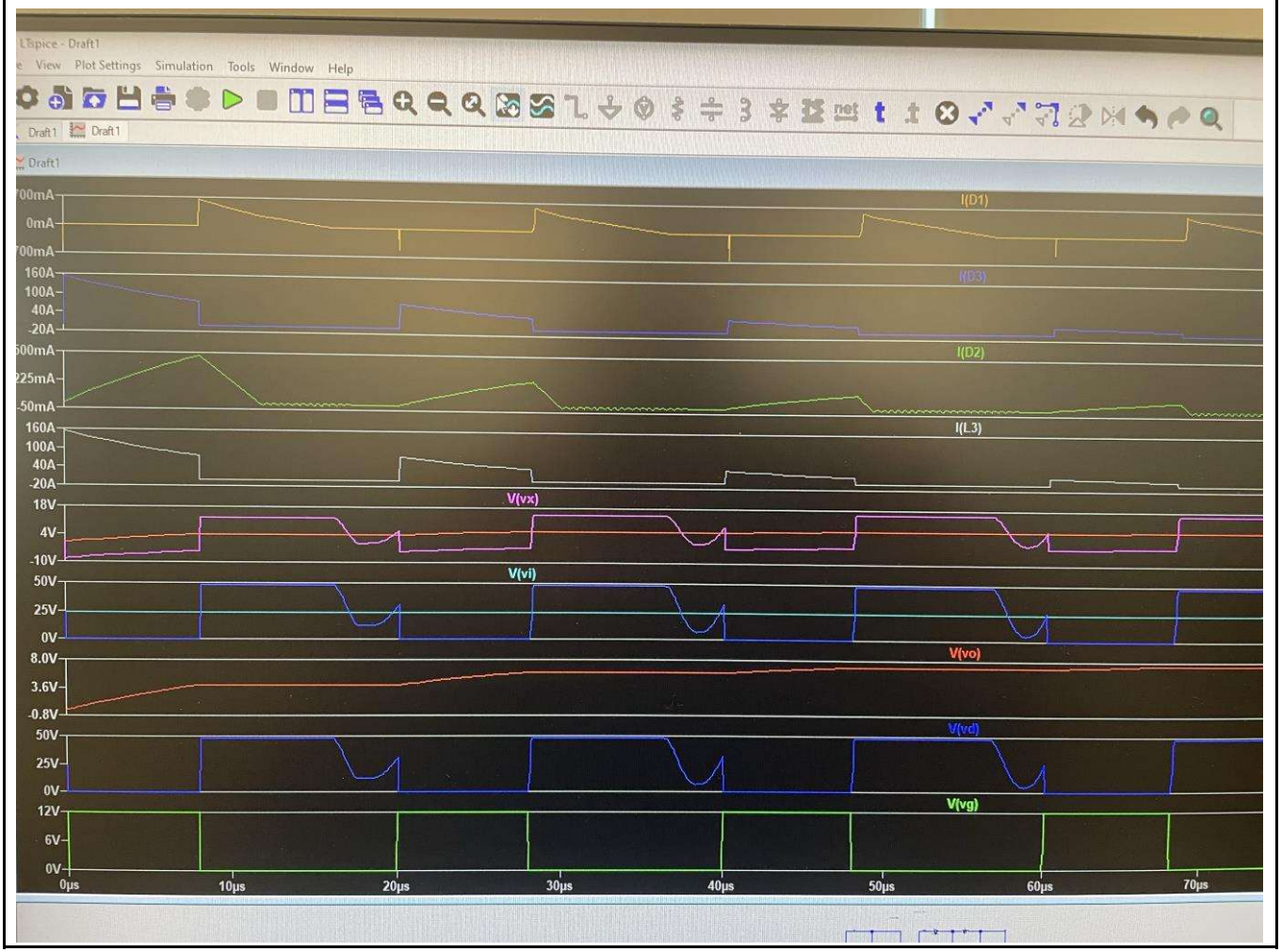
4) Primary winding voltage V(Vi,Vd) 5) Output-inductor voltage V(Vx,Vo).

6) Output-inductor current I(L4).

7) Rectifier diode current I(D2) 8) Freewheeling diode current I(D3).

9) Reset diode current I(D1).

Show your Forward converter waveform screenshot:



Measured reset duration = ____8_____ µs; OFF time = ____12_____ µs.

Does I(D1) return to zero before the next ON interval? __Yes_____.

4. Complete the table using your waveforms.

Quantity	Switch ON	Switch OFF
Rectifier D2 conducting? (Yes/No)	Yes	No

Freewheeling D3 conducting? (Yes/No)	No	Yes
Output-inductor voltage V(Vx,Vo)	7.2 V	-4.8 V
Output-inductor current rises or falls?	Rises	Falls
Current path supplying the load	Secondary -> D2 -> L4 -> load/Co	L4 -> load -> D3

Use the ideal CCM equations below. Forward CCM refers to the output-inductor current I(Lo). The transformer must still reset each cycle.

$$V_o = ndV_{in} \qquad I_o = \frac{V_o}{R_L}$$

$$\Delta i_{Lo} = \frac{(nV_{in} - V_o)d\Delta i_{Lo}}{L_o f_s 2} \qquad I_{Lo,min} = I_o - \frac{}{}$$

$$\Delta V_o \approx \frac{}{8 f_s C_o} \qquad V_{D,reset} = V_{in}\left(1 + \frac{\Delta i_{Lo} N_1}{}\right) \frac{}{N_3}$$

Quantity	Calculated	Simulated
Average V_o (V)	4.8 V	4.5 V
Output-inductor ripple Δi_{Lo} (A, p-p)	1.152 A	1.1 A
Output ripple ΔV_o (V, p-p)	13.1 mV	13 mV
Switch voltage during reset $V_{D,reset}$ (V)	48 V	47.2 V

This linear transformer model cannot predict core saturation. In a real converter, incomplete reset can cause flux accumulation and saturation!

Q: A forward converter can be viewed as an isolated version of a buck converter. What are the benefits of introducing isolation into the DC-DC converter?

It provides safety and breaks ground loops, which can preserve equipment and prevent faults.

Compare the Flyback and Forward converters:

Characteristic	Flyback	Forward
Main energy-storage element	Transformer magnetizing inductance	Output inductor L4
When energy is transferred through the secondary (MOSFET ON or OFF)	OFF	ON
Ideal CCM voltage gain $\frac{V}{V_{in}}$	n*d / (1-d)	n*d
Transformer reset necessary?	No	Yes

Q: How does the role of the two isolated inductor windings (transformer windings) in a flyback converter differ from that of the two isolated windings in a forward converter?

In the flyback, the two windings act as a single coupled inductor, while in a forward converter, they act as a true transformer. In the flyback, the primary stores energy while M1 is on, and the secondary releases it while M1 is off, so they never conduct at the same time. In the forward converter, both conduct together during on and pass power straight to the output, and the third winding is there only to reset the magnetizing flux.

Summary

Write a brief summary of what you have learned from this lab.

Today we simulated a flyback convertor in CCM and DCM as well as a forward convertor. When simulating the flyback convertor, we found the measured waveform values aligned closely with the calculated values we found, showing that we correctly operated it

in both CCM and DCM. We also learned about the forward convertor and successfully simulated its function, with values calculated values with 10% of the measured values.

Before leaving

Save all LTspice schematics and your completed lab sheet. Include the requested waveform screenshots and measurement tables.